Ishitha …there are 2 prompts …keep scrolling …if one doesn't work others will

Got it — you want **literally everything from zero to deployment**, step by step. I'll walk you through like a lab manual so you can follow blindly and still finish it perfectly.

🔵 PART 1: Create Project Folder

1. Create a folder

- Right click → New Folder
- Name it:

university-department-webpage

2. Open in VS Code

- Open VS Code
- Click **File** → **Open Folder** → **select this folder**

🔵 PART 2: Create Required Files

Inside VS Code:

Create these files:

index.html
style.css
Dockerfile
Jenkinsfile
README.md

🔵 PART 3: Write HTML Code

Open `index.html` → paste this FULL code:

```
<!DOCTYPE html>
<html>
<head>
   <title>Computer Science Department</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>

<header>
   <h1>ABC University</h1>
   <p>Welcome to the Computer Science Department</p>
   <p>Head of Department: Dr. John Doe</p>
</header>

<section>
   <h2>About Department</h2>
   <p>The Computer Science Department offers quality education in software, AI, and data
science.</p>
</section>

<section>
   <h2>Courses Offered</h2>
   <ul>
      <li><b>Artificial Intelligence</b> - Basics of AI and machine learning.</li>
      <li><b>Data Structures</b> - Study of data organization.</li>
      <li><b>Web Development</b> - Learn HTML, CSS, JavaScript.</li>
   </ul>
</section>

<section>
   <h2>Faculty</h2>
   <p>Dr. John Doe - Professor (AI)</p>
   <p>Dr. Jane Smith - Associate Professor (Data Science)</p>
</section>

<section>
   <h2>Contact</h2>
   <p>Address: Chennai, India</p>
   <p>Phone: 1234567890</p>
   <p>Email: cs@abcuniversity.com</p>
</section>

<footer>
```

```
    <p>© 2026 ABC University</p>
</footer>

</body>
</html>
```

🔵 PART 4: Add CSS Styling

Open `style.css` → paste:

```
body {
    font-family: Arial;
    margin: 0;
    padding: 0;
}

header {
    background-color: #2c3e50;
    color: white;
    text-align: center;
    padding: 20px;
}

section {
    padding: 20px;
}

footer {
    background-color: #2c3e50;
    color: white;
    text-align: center;
    padding: 10px;
}
```

🔵 PART 5: Run Website

Option 1 (easy):

- Install **Live Server** extension in VS Code
- Right click `index.html` → **Open with Live Server**

🔵 PART 6: GitHub Setup

1. Initialize Git

Open terminal in VS Code:

```
git init
```

2. Add files

```
git add .
```

3. Commit

```
git commit -m "Initial commit"
```

---

4. Create GitHub Repo

- Go to GitHub
- Click **New Repository**
- Name:

university-department-webpage

---

5. Connect Repo

```
git remote add origin
https://github.com/YOUR_USERNAME/university-department-webpage.git
git branch -M main
git push -u origin main
```

🔵 PART 7: Clone Repo (as required in question)

```
git clone https://github.com/YOUR_USERNAME/university-department-webpage.git
cd university-department-webpage
```

🔵 PART 8: Docker Setup

Create `Dockerfile`

Paste this:

```
FROM nginx:alpine

COPY . /usr/share/nginx/html

EXPOSE 80
```

Build Docker Image

```
docker build -t university-webpage .
```

Run Container

```
docker run -d -p 8080:80 university-webpage
```

Open browser:

http://localhost:8080

🔵 PART 9: Push Docker Image (Optional but good)

```
docker tag university-webpage YOUR_USERNAME/university-webpage
docker push YOUR_USERNAME/university-webpage
```

🔵 PART 10: Jenkins Setup

Install Jenkins (if not already)

Use Docker:

docker run -p 8081:8080 jenkins/jenkins:lts

Create Pipeline Job

- Open: `http://localhost:8081`
- New Item → Pipeline

Create `Jenkinsfile`

Paste:

```
pipeline {
    agent any

    stages {
        stage('Clone Repo') {
            steps {
                git 'https://github.com/YOUR_USERNAME/university-department-webpage.git'
            }
        }

        stage('Build Docker Image') {
            steps {
                sh 'docker build -t university-webpage .'
            }
        }

        stage('Run Container') {
            steps {
                sh 'docker run -d -p 8082:80 university-webpage'
            }
        }
    }
}
```

🔵 PART 11: Kubernetes (Basic Setup)

Create `deployment.yaml`

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: university-web
spec:
  replicas: 1
  selector:
    matchLabels:
      app: web
  template:
    metadata:
      labels:
        app: web
    spec:
      containers:
      - name: web
        image: YOUR_USERNAME/university-webpage
        ports:
        - containerPort: 80
```

Run in Kubernetes

```
kubectl apply -f deployment.yaml
```

🔵 PART 12: Final Checklist (IMPORTANT)

✔ HTML page created
✔ CSS styling done
✔ GitHub repo created
✔ Code pushed
✔ Repo cloned
✔ Docker container running
✔ Jenkins pipeline created
✔ Kubernetes deployment done

🔥 If You Want Extra Marks

Add:

- Navigation bar
- Images
- Better CSS
- Responsive design

If you want, I can:
✅ Convert this into **perfect PDF report**
✅ Add **screenshots + viva answers**
✅ Or simplify Jenkins/Kubernetes (if your lab expects less)

Just tell me 👍

PROMPT 2:

Alright — this time I'll give you a **clean, Windows-focused, no-missing-steps master guide** you can literally follow during your lab without thinking. Nothing skipped, nothing assumed.

🟢 PHASE 0: One-Time Windows Setup (DO THIS FIRST)

1. Install Software

Install these:

- Visual Studio Code
- Git
- Docker Desktop

👉 While installing Git:

- Select **"Git Bash Here"**
- Select **"Use Git from command line"**

---

2. Enable WSL (VERY IMPORTANT)

Open **PowerShell as Administrator**:

wsl --install

Restart your laptop.

---

3. Start Docker Desktop

- Open Docker Desktop manually
- Wait till it says **"Running"** (don't skip this ever)

🟢 PHASE 1: Create Project

Step 1: Create Folder

On Desktop:

university-department-webpage

Step 2: Open in VS Code

- Open VS Code
- File → Open Folder → select your folder

Step 3: Create Files

Inside VS Code create:

index.html
style.css
Dockerfile
Jenkinsfile
deployment.yaml

README.md

🟢 PHASE 2: Write Code

1. index.html

Paste:

```
<!DOCTYPE html>
<html>
<head>
   <title>Computer Science Department</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>

<header>
   <h1>ABC University</h1>
   <p>Computer Science Department</p>
   <p>Head: Dr. John Doe</p>
</header>

<section>
   <h2>About</h2>
   <p>This department focuses on AI, ML and Software Engineering.</p>
</section>

<section>
   <h2>Courses</h2>
   <ul>
      <li>AI - Machine learning basics</li>
      <li>Data Structures - Data organization</li>
      <li>Web Dev - HTML CSS JS</li>
   </ul>
</section>

<section>
   <h2>Faculty</h2>
   <p>Dr John Doe - AI</p>
   <p>Dr Jane Smith - Data Science</p>
</section>

<section>
```

```
    <h2>Contact</h2>
    <p>Chennai, India</p>
    <p>1234567890</p>
    <p>cs@abc.com</p>
</section>

<footer>
    <p>© 2026 ABC University</p>
</footer>

</body>
</html>
```

---

2. style.css

```
body {
    font-family: Arial;
    margin: 0;
}

header {
    background: #2c3e50;
    color: white;
    text-align: center;
    padding: 20px;
}

section {
    padding: 20px;
}

footer {
    background: #2c3e50;
    color: white;
    text-align: center;
    padding: 10px;
}
```

🟢 PHASE 3: Run Website

Install Live Server (VS Code)

- Extensions → search → Live Server → install

Run:

- Right click `index.html` → **Open with Live Server**

🟢 PHASE 4: Git + GitHub

Step 1: Open Terminal

👉 IMPORTANT: Use **Git Bash** (not CMD)

In VS Code:

- Terminal → New Terminal → select **Git Bash**

Step 2: Initialize Git

```
git init
git add .
git commit -m "first commit"
```

Step 3: Create GitHub Repo

Go to GitHub → New repo
Name:

university-department-webpage

Step 4: Connect Repo

```
git remote add origin
https://github.com/YOUR_USERNAME/university-department-webpage.git
git branch -M main
git push -u origin main
```

Step 5: Clone (as question asks)

```
git clone https://github.com/YOUR_USERNAME/university-department-webpage.git
cd university-department-webpage
```

🟢 PHASE 5: Docker

Step 1: Create Dockerfile

```
FROM nginx:alpine
COPY . /usr/share/nginx/html
EXPOSE 80
```

Step 2: Build Image

```
docker build -t university-webpage .
```

Step 3: Run Container

```
docker run -d -p 8080:80 university-webpage
```

Step 4: Open in Browser

http://localhost:8080

👉 If not working → change port:

```
docker run -d -p 9090:80 university-webpage
```

🟢 PHASE 6: Jenkins (Using Docker)

Step 1: Run Jenkins

docker run -p 8081:8080 jenkins/jenkins:lts

Step 2: Open Jenkins

http://localhost:8081

Step 3: Create Pipeline

- New Item → Pipeline
- Name: `web-pipeline`

Step 4: Jenkinsfile

Paste:

```
pipeline {
    agent any

    stages {
        stage('Clone') {
            steps {
                git 'https://github.com/YOUR_USERNAME/university-department-webpage.git'
            }
        }

        stage('Build Docker') {
            steps {
                sh 'docker build -t university-webpage .'
            }
        }

        stage('Run Docker') {
            steps {
                sh 'docker run -d -p 8082:80 university-webpage'
            }
        }
    }
}
```

🟢 PHASE 7: Kubernetes (via Docker Desktop)

Step 1: Enable Kubernetes

- Open Docker Desktop
- Settings → Kubernetes → Enable

Step 2: deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: university-web
spec:
  replicas: 1
  selector:
    matchLabels:
      app: web
  template:
    metadata:
      labels:
        app: web
    spec:
      containers:
      - name: web
        image: YOUR_USERNAME/university-webpage
        ports:
        - containerPort: 80
```

Step 3: Run

```
kubectl apply -f deployment.yaml
```

🟢 PHASE 8: MOST COMMON WINDOWS ERRORS (READ THIS)

❌ Docker not working

✔ Start Docker Desktop manually

❌ Port already in use

✔ Change port (8080 → 9090)

❌ Command not found

✔ Use Git Bash instead of CMD

❌ Path issues

✔ Use:

/c/Users/YourName/project

❌ Permission issues

✔ Run terminal as Administrator

🟢 FINAL CHECKLIST

✔ HTML + CSS done
✔ Website runs
✔ GitHub repo pushed
✔ Repo cloned
✔ Docker image built
✔ Container running
✔ Jenkins pipeline created
✔ Kubernetes deployed

🔥 If you want to be fully exam-ready

I can next give you:

- **Exact viva answers (word-to-word)**
- **What to say while demonstrating**
- **How to explain each step confidently**

Just tell me 👍